MATH3881/5881: Statistical Machine Learning Theory

Week 1: Tutorial Problems — Introduction to Machine Learning

Sarat Moka *UNSW, Sydney*

These three problems accompany the Week 1 lecture notes (*Introduction to Machine Learning*), covering *empirical risk minimisation*, *linear regression*, *logistic regression*, and *gradient descent*. Attempt them with pen and paper before the tutorial; a separate solutions document is provided a week after the tutorial session. A companion Python notebook provides lab practice on the same ideas.

Problem 1: From ERM to the least-squares estimates

Consider supervised learning with one-dimensional inputs $x^{(i)} \in \mathbb{R}$ and continuous targets $y^{(i)} \in \mathbb{R}$, $i = 1, \ldots, n$.

(a) Start with the simplest possible model, the *constant predictor* $f_\theta(x) = \theta_0$. Under the squared loss the empirical risk is

$$C(\theta_0) \;=\; \frac{1}{n}\sum_{i=1}^{n}\big(y^{(i)} - \theta_0\big)^2.$$

Show that its minimiser, known as the least-squares estimate, is the sample mean $\hat{\theta}_0 = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y^{(i)}$.

(b) Now take the *simple linear regression* model $f_\theta(x) = \theta_0 + \theta_1 x$ with the squared loss, so that

$$C(\theta_0, \theta_1) \;=\; \frac{1}{n}\sum_{i=1}^{n}\big(y^{(i)} - \theta_0 - \theta_1 x^{(i)}\big)^2.$$

By setting both partial derivatives of C to zero, derive

$$\hat{\theta}_1 \;=\; \frac{\sum_{i=1}^{n}(x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^{n}(x^{(i)} - \bar{x})^2}, \qquad \hat{\theta}_0 \;=\; \bar{y} - \hat{\theta}_1\,\bar{x}.$$

(c) Define the residual at the i-th point as $r^{(i)} = y^{(i)} - (\hat{\theta}_0 + \hat{\theta}_1 x^{(i)})$. Show that the residuals sum to zero:

$$\sum_{i=1}^{n} r^{(i)} \;=\; 0.$$

(*Hint:* this is the normal equation $\partial C/\partial\theta_0 = 0$ in disguise.) Use this to argue that the fitted line always passes through the data's "centre of mass" $(\bar{x}, \bar{y})$.

Problem 2: Probabilities and the decision boundary

A logistic regression model with $p = 2$ features has been fitted, giving parameter estimates $\hat{\theta} = (\hat{\theta}_0, \hat{\theta}_1, \hat{\theta}_2) = (-1,\ 2,\ -1)$, so that the predicted probability is

$$\hat{y} \;=\; \sigma\big(\hat{\theta}_0 + \hat{\theta}_1 x_1 + \hat{\theta}_2 x_2\big) \;=\; \sigma\big(-1 + 2x_1 - x_2\big), \qquad \text{where } \sigma(z) = \frac{1}{1+e^{-z}}.$$

(a) Compute the predicted probability $\hat{y}$ at the test points $x = (1, 1)$ and $x = (2, 1)$.

(b) Predict the class label of each point using the decision threshold $\tau = 0.5$ (predict class 1 if $\hat{y} > \tau$). Then repeat using $\tau = 0.3$. (*Hint:* since σ is strictly increasing, $\hat{y} > \tau \iff \hat{\theta}_0 + \hat{\theta}_1 x_1 + \hat{\theta}_2 x_2 > \log \frac{\tau}{1-\tau}$.)

(c) Show that the decision boundary $\{x : \hat{y} = \tau\}$ is the line

$$\hat{\theta}_0 + \hat{\theta}_1 x_1 + \hat{\theta}_2 x_2 \;=\; \log \frac{\tau}{1-\tau},$$

and write it explicitly for $\tau = 0.5$ and for $\tau = 0.3$. Explain why decreasing τ shifts the decision boundary line *parallel* to itself (same slope), classifying more points as positive.

Problem 3: One step of gradient descent by hand

Train a logistic regression model ($p = 1$, augmented input $\tilde{x} = (1, x)^\top$) on the two points $\big(x^{(1)}, y^{(1)}\big) = (1, 1)$, $\big(x^{(2)}, y^{(2)}\big) = (-1, 0)$. Start from $\theta^{(0)} = (0, 0)^\top$ with learning rate $\alpha = 1$. Recall from the lecture notes that the gradient of the binary cross-entropy loss is

$$\nabla_\theta C(\theta) \;=\; \frac{1}{n} X^\top \big(\hat{y} - y\big), \qquad \hat{y} = \sigma(X\theta),$$

where X is the design matrix with rows $\tilde{x}^{(i)\top}$.

(a) Write down X and y explicitly. Compute the predictions $\hat{y}^{(0)}$ at $\theta^{(0)}$ and the initial cost $C(\theta^{(0)})$.

(b) Compute the gradient $\nabla_\theta C(\theta^{(0)})$ and the updated parameters $\theta^{(1)} = \theta^{(0)} - \alpha\, \nabla_\theta C(\theta^{(0)})$.

(c) Look only at the gradient $\nabla_\theta C(\theta^{(0)})$ you computed in part (b). Without re-using the update formula, decide — and briefly justify in one sentence each — in which direction you would change each coordinate of θ to reduce the cost C:

- The intercept θ_0: *increase*, *decrease*, or *hold fixed*?
- The slope θ_1: *increase*, *decrease*, or *hold fixed*?

Check that your answers agree with the update $\theta^{(1)}$ from part (b).